Ryan Charles Neu

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EDUCATION

Villanova University, College of Engineering — Expected May 2028
Bachelor of Science in Mechanical Engineering — GPA: 3.92
Minors: Mechatronics, Business (Villanova School of Business)
Honors: Dean's List (2024 – Present) | Pi Tau Sigma (Mechanical Engineering Honor Society)
Relevant Coursework: Dynamic Systems II, Solid Mechanics, Heat Transfer, Computer-Aided Design, Programming for ME, Circuits

ENGINEERING EXPERIENCE AND PROJECTS

Systems Engineering Lead & Team Lead – INNOVATE: L3Harris Endowed Summer Program | Villanova University — Summer 2026

- Conceived and led development of LiveLine, a real-time power-line fault monitoring prototype, directing a 4-person cross-disciplinary team through mechanical design, sensor integration, and fabrication to deliver a functional 5-sensor prototype
- Designed sensor integration architecture, selecting, positioning and mounting 5 sensors to detect 12 fault-relevant metrics while defining structural mounting requirements across an aluminum clamp and PLA enclosure
- Verified all 5 sensor subsystems functional through bench testing (strain, IMU, acoustic, temperature, current), validating current sensor accuracy within 10% against a watt-meter reference

Undergraduate Research Assistant – Autonomous Rover Research | Villanova University — October 2025 – Present

- Designed a wheeled rover in SolidWorks, sizing motors and drivetrain from weight/torque calculations for towing other robots
- Machined the aluminum chassis, wired and configured ESCs, and validated the drivetrain under RC control; redesigned the wheel drive with a machined drive shaft and outboard bearing to fix a shaft-fatigue risk, raising calculated FOS from 0.9 to over 3
- Built a ROS 2/Gazebo simulated rover with lidar, slam_toolbox mapping, Nav2 waypoint navigation, and Python patrol scripts
- Calibrated a RadioLink drone with ArduPilot and delivered a multi-robot coordination demonstration to the Army Research Lab

Engineering Lead & Team Lead – Villanova Combat Robotics | Villanova University — January 2025 – Present

- Designed and manufactured the full weapon system for FREEDOM, a 3-lb vertical spinner, from CAD through 4140 steel fabrication on a lathe and mill; validated the weapon and dead-shaft assembly against impact loading using FEA
- Fielded FREEDOM at the National Havoc Robotics League (NHRL), Villanova Combat Robotics' first-ever competition entry
- Promoted to 2026–2027 Engineering and Team Lead for the robot build, directing mechanical design, systems integration, build scheduling across subsystems, developing new-member training curriculum, continuing to lead weapon system development

Project Manager and Chassis Design Lead – Villanova Beetle Bot Competition | Villanova University — Fall 2025

- Managed a 4-person team, setting deadlines, assigning roles, and tracking progress to deliver all project milestones on schedule
- Engineered a 3-lb robot chassis in SolidWorks, iterating mechanical designs to optimize maneuverability and pushing power
- Competed robot in Villanova's Beetle Bot competition and won two of the three matches against opposing student teams

SKILLS

Software & CAD: SolidWorks (CAD/FEA), GD&T (ASME Y14.5), Python, MATLAB, ROS 2, Nav2, Gazebo, Arduino IDE, Mission Planner
Fabrication & Prototyping: Machine Shop Certified (Lathe, Mill, Drill Press, Bandsaw), 3D Printing, Electronics Integration, Soldering
Skills In Development: ANSYS (FEA), C++, EKF State Estimation, NVIDIA Isaac Sim, Simulink

LEADERSHIP & INVOLVEMENT

Resident Assistant – Villanova Residence Life | Villanova University — August 2025 – Present

• Selected as RA for 2 consecutive years (first-year hall, then upperclassmen apartments); completed 50+ hours of intensive leadership training, developed and executed community-building programs and mentorship initiatives while managing a project budget

VUnited Scholars Program Living and Leisure Mentor – VUnited | Villanova University — January 2025 – Present

• Mentored 25 students with intellectual disabilities, supporting academic, social, and independent-living skill development

ASME and NSPE, *Member* | Villanova University — September 2024 – Present